A Theory of Escalation and International Conflict

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Escalation processes are found in many types of international conflict. However, a great deal of the theoretical and empirical literature on escalation is context specific and concentrates on explaining the outcomes of an escalation process. This approach has generated numerous insights; however, our understanding of escalation processes, in general, remains partial and incomplete. In this article, the author develops a two-sided incomplete information model to identify the kinds of escalation strategies states are likely to adopt in conflict. The model produces several hypotheses, one of which is tested empirically in the context of militarized interstate disputes. The hypothesis states that as the disparity between the players' cost tolerances increases, the lower cost tolerant actor is more likely to escalate to the maximum of his or her ability on the first move in the conflict. The results of the test confirm the theory's expectations of an inverse relationship between cost tolerance and an actor's escalation behavior. The article concludes by noting implications for future research on escalation processes.

The recent debate within the North Atlantic Treaty Organization (NATO) over what strategy to pursue toward the war-torn former Yugoslavia provides a poignant example of the passionate reactions evoked by the mere suggestion of "escalation." The NATO debate exposes a serious problem for both policymakers and international relations scholars dedicated to understanding escalation processes. Decision makers often confront a dilemma in trying to minimize the costs that must be paid to achieve their goal in conflict. Any attempt to analyze the likelihood that a state will prevail in a conflict must be guided by a theoretical framework that specifies the relationships among key variables affecting escalation decisions.

The problem of escalation has been widely discussed (Kahn 1965; Schelling 1960, 1966; Holsti 1972; Smoke 1977; Brecher 1994), but until recently there have been few attempts to develop a general theory aimed at

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generating testable hypotheses of escalation processes. In the absence of empirical testing, it is difficult to assess the comparative explanatory and predictive capabilities of competing escalation models. Of those studies that have focused on escalation processes in general, many are wedded conceptually to a particular type of international context such as arms races (Richardson 1960; Intriligator and Brito 1984) or deterrence situations (Brams and Kilgour 1987, 1988; Zagare 1987, 1992). These studies have contributed to our knowledge of particular processes, but our understanding remains partial and incomplete.

The purpose of this article is to present a theory of escalation processes. The model is based on the theory of games and is used to derive hypotheses identifying the conditions under which states escalate and the level of escalation states achieve during conflict. The model is sufficiently general to support hypotheses in a variety of contexts in which escalation can occur. These contexts include arms races, crisis bargaining, deterrence, and war. In this article, one of the model's hypotheses is tested empirically in the context of militarized interstate disputes (Gochman and Maoz 1984).

More specifically, the hypothesis states that as the disparity between the players' cost tolerances increases, there is an increase in the probability that the lower cost tolerant (LCT) actor achieves his or her maximum level of escalation at the outset of the conflict. The results of the empirical test support the theory's expectations and suggest that further research is warranted.

The article is organized as follows. The first section examines the previous theoretical and empirical efforts to understand escalation processes. The second section develops the model used to identify the levels of escalation states are likely to achieve in conflict. The subsequent section discusses the research design used to test the hypothesis derived from the model. This is followed by an analysis of the test results. The final section contains concluding remarks.

LITERATURE REVIEW

Escalation processes lie at the heart of many contexts of state interaction. Escalation has been studied in connection with arms races (Richardson 1960; Intriligator and Brito 1984), deterrence (Brams and Kilgour 1987; Zagare 1992), and escalation in war (Smoke 1977; Wittman 1979; Pillar 1983). International crises have been an especially important focal point for research on escalation processes (Schelling 1960, 1966; Holsti 1972; Lebow 1981; Leng 1980, 1993; Siverson and Miller 1993; Brecher 1994). In a crisis situation, not only must states decide whether to escalate, but actors also must

consider the degree of coercive pressure to direct against an opponent (Lockhart 1979). Although there has been a great deal of attention paid to the factors that affect a state's decision to escalate to the highest levels, there are fewer systematic, theoretical, and empirical studies identifying the level of escalation states achieve short of war.

Many of the earliest attempts to address the question of escalation placed the bargaining process at center stage (Schelling 1960, 1966; Kahn 1965; Young 1968; Snyder 1972; Smoke 1977). According to Schelling (1960, 1966), one of the potential effects of escalation is to convince an opponent to back down by exploiting his or her fear that future escalation will lead to disaster. Because both actors are engaged in demonstrating their superior ability to tolerate these risks, escalation is conceptualized as a game of competitive risk taking (Schelling 1960; Kahn 1965; Powell 1987, 1988; Maoz 1985, 1990; Geller 1990). Although Kahn (1965) and others have recognized that the rate at which states escalate (impose costs) can have an important effect on the bargaining process, few have specified, theoretically, the conditions under which actors select different levels of escalation to achieve their ends during the bargaining process.

Some case studies on escalation have dealt more directly with identifying the kinds of escalation strategies states adopt in conflict (Ikle 1971; Smoke 1977; Snyder and Diesing 1977; Pillar 1983; Haig 1984). In general, the discussion revolves around whether a series of small escalation steps (low escalation) or a few decisive escalation moves (high escalation) is more likely to produce an opponent's concession. States that use a series of small escalation actions minimize their relative costs; however, this option runs the risk of counterescalation by failing to isolate the opponent's tolerance for escalation (Haig 1984; Patchen 1988). On the other hand, some argue that high escalation offers a better opportunity to surpass the opponent's cost threshold, but the price of that concession may exceed what was necessary to secure the outcome (Bonoma 1975).

There is by no means any consistent theoretical agreement on these propositions, however. Synder and Diesing (1977) have argued that, under certain conditions, high escalation is more likely to produce counterescalation than it is to produce a concession. In addition, it has been argued that low, gradual escalation is more likely if states have greater resources than does an opponent (Brecher 1994) and that low escalation works for states that have fewer resources and for whom the issue in dispute is less important (Lockhart 1979). Whereas part of the inconsistency in findings is due to the fact that research in this area is guided by different assumptions and different theoretical frameworks, one of the more serious problems is that few of these debates are imbedded within a theoretical framework that rigorously speci-

fies how the key variables interact and what conditions lead states to select among the various levels of escalation.

There does exist fairly widespread agreement that actors' relative military capabilities and the value they attach to the issue at stake are two of the key variables affecting the decision to escalate (Leng and Gochman 1982; Gochman and Maoz 1984; Siverson and Miller 1993). Military capabilities are clearly important because they function as the primary means of imposing costs on an opponent during a crisis. Despite persistent theoretical disagreements (Brecher 1994), there is a consistent body of empirical evidence indicating that states of relatively equal power are more likely to go to war with one another than are states with disparate capabilities (Bremer 1992). There is also empirical evidence suggesting that actors who place a greater value on the issue in dispute are more willing to escalate to the highest levels to achieve a demand (Gochman and Leng 1983; Maoz 1983; Wilkenfeld 1991; Brecher 1994).

These studies have been concerned primarily with explaining the conditions under which states escalate to the extreme (i.e., war). How these variables interact at levels short of war to affect a state's choice over escalation strategies remains unclear. Recent attempts to understand bargaining and/or escalation processes with formal models have begun to shed light on why weaker or unresolved actors do not always capitulate immediately to the stronger side's demands (Powell 1987; Morrow 1989; Lalman 1990; Zagare 1992). Under certain conditions, a stronger actor can misperceive his or her bargaining position and give in, or an unresolved actor can achieve his or her demand by bluffing a willingness to bear the risks of escalation.

Even though this literature does not explicitly identify the theoretical interaction between capabilities/values and a state's decision to use a low or high escalation strategy short of war, it is clear that a player's optimal escalation strategy is conditioned by his or her perceptions of the opponent's capabilities and value over the issue at stake. Moreover, by manipulating the rate of escalation, a state can attempt to send an opponent a signal regarding the willingness to bear escalation costs.

Presently, however, there exists a large gap in our theoretical and empirical treatments of escalation. Formal models of escalation processes have produced a great deal of insight on how uncertainty over an opponent's willingness to bear costs and the evaluation of a demand affects escalation decisions. Although there exists a general consensus on some of the key variables affecting states' escalation behavior in both the theoretical and empirical literature, there is also a noticable lack of agreement on how these variables interact to produce low- or high-escalation strategies short of war. The

remainder of this article seeks to fill the gap between the theoretical and empirical research on this facet of escalation.

A THEORY OF ESCALATION

Escalation is conceptualized as a tool states use during bargaining (Schelling 1960, 1966; Kahn 1965; Young 1968; Snyder 1972). The bargaining process is initiated by one or both actors to reconcile their positions over some issue in dispute. Actors begin the bargaining process by articulating their proposals for a negotiated solution, which can be either accepted or rejected by the opposing player. If one side accepts the terms for agreement, then actors receive the benefit they associate with the proposed outcome. If an offer is rejected, then the bargaining process continues and neither player receives any benefits until one of the sides concedes to a demand.[1] The problem facing both actors is to provide the opponent with an incentive to make concessions. Actors can create that incentive by making it costly for the opponent to persist in rejecting demands (Cross 1969; Pillar 1983; Morgan 1990, 1994). Escalation is one way that actors can make bargaining costly.

Escalation imposes costs on both actors. For instance, the escalating state suffers the costs associated with expending resources and risking lives, whereas the receiving state suffers the costs of rebuilding industrial centers and the like. The benefit of a negotiated outcome decreases for both actors when one of them decides to escalate. As the costs associated with disagreement (or the value of a settlement) increase, players are more likely to give in to their opponent's demand. An actor concedes if the costs of enduring further escalation will outweigh the benefits of giving in to the opponent. Thus, as Schelling (1966) points out, escalation is the coercive side of bargaining in which the fear of even greater cost imposition motivates states to concede.

The maximum escalation costs that a player will endure to achieve a demand is referred to as a player's cost tolerance. I assume that the price that an actor is willing to pay to play depends on how an actor evaluates the

1. The conceptualization of bargaining offered here differs from many economic models of the bargaining process. For example, Rubenstein (1982) conceives bargaining as a problem over how to divide a pie between two players. Players bargain until an offer is accepted by one side. In my conceptualization of the bargaining process, the players are not concerned with distributing the gains from a pie. First, players may or may not agree on the size of the pie. Second, the players' bargaining behavior is aimed at convincing the opponent to concede to an entire demand.

importance of the issue at stake. A player who considers the issue at stake to be of greater salience than some other issue that might arise should be willing to pay greater costs to secure the opponent's concession. Conversely, a player will tolerate fewer costs to achieve a demand when the issue at stake is deemed relatively less important. Thus a player will continue to bargain only if he or she believes that the opponent's cost tolerance is lower than his or her own.

To illustrate this idea, consider a situation in which both players know the opponent's cost tolerance (players always know their own cost tolerance). The prediction is that the LCT player concedes immediately. The LCT player knows he or she will lose eventually to the higher cost tolerant (HCT) opponent. Thus the former concedes immediately to avoid paying escalation costs. It follows that escalation is possible only when players are uncertain about key aspects of their strategic environment (Wagner 1982; Powell 1987, 1989; Lalman 1990). For example, if President Lyndon Johnson would have known the outcome of the Vietnam War, he would not have escalated American involvement beginning in 1964.

An actor escalates bargaining under conditions of incomplete information to project the image that he or she has a higher cost tolerance than does the opponent. The problem a player confronts is to determine whether the observed escalation behavior provides an accurate reflection of the opponent's true cost tolerance. A player's ability to draw the appropriate conclusion is complicated by the fact that players of varying cost tolerances are motivated to escalate at different levels depending on their beliefs about the adversary.

All players are motivated to minimize the costs necessary to achieve a demand. These costs can vary depending on the particular mix of cost tolerant actors engaged in a dispute. HCT actors may have an incentive to escalate at lower levels early on in the conflict. If the opponent backs down, the HCT actor obtains a greater net benefit than what would have resulted from escalating at higher levels. On the other hand, if the opponent refuses to concede, HCT players are willing to pay additional escalation costs during the next round (Ikle 1971; Patchen 1988). LCT players can also play that type of escalation game, especially if they perceive that their competitor is also LCT. However, if the opponent is perceived to be HCT, then a long series of graduated escalation moves might not convince the opponent to give in. In particular, escalating at higher levels from the outset of the conflict might lead an opponent to believe that the escalating player has a higher cost tolerance than is actually the case. Thus there is some chance that an LCT actor can achieve a demand by manipulating the costs of escalation. The

question for the LCT player is, Which level of escalation maximizes his or her expected utility?

To analyze this question, a two-sided incomplete information model is developed. There are two actors, states i and j. Each player has the choice of escalating low or high.[2] The alternatives differ in terms of the costs generated by an actor for carrying out that action (e.g., a higher level of escalation equals higher cost imposition). I assume that players are able to discern the cost differential associated with each escalation option. Thus players can determine the choice made by the opponent. I assume that nature selects player i to move first and last in the game. Player j has a single choice of escalating low or high.[3]

The key feature of the model is that i and j can be one of two types: LCT or HCT.[4] I assume that LCT actors will withstand one high-escalation action, taken by either actor, to achieve a demand. If high escalation fails to convince the opponent to back down, then LCT players would rather give in than suffer additional costs over an issue deemed less important than some other issue that might be contested. On the other hand, HCT actors will endure two high-escalation actions to achieve a demand. The willingness of each type to suffer different costs is modeled by distinct preference orderings over the outcomes. I return to this point in the following section.

Both players know their own type but are uncertain of the opponent's true type. Player i (j) knows only the distribution of types in the population of possible j (i) players and the preference ordering associated with each j (i) type. These distributions serve as the players' prior probabilities that the opponent is a particular type. Player i's escalation decision on the first move serves as an input or additional information that j uses to revise his or her prior estimates of i's type. I assume that j uses Bayesian updating to revise his or her prior beliefs regarding i's cost tolerance. Player j uses his or her updated beliefs about i's type to determine whether he or she should escalate low or high. Player i has the final move, and each i type selects the alternative that produces the outcome with the greater expected utility.

2. A more realistic assumption is that the level of escalation is continuous. Allowing the players to choose between two escalation options provides an extremely useful first cut for identifying the conditions under which the players escalate to one of two clearly defined levels.

3. The model could include another move for player j. In fact, the game could be extended indefinitely. However, the gain in descriptive accuracy is bought at the price of reducing the analytic rigor of the model. The calculation to determine the equilibrium outcome after including an additional move would be identical to the previous calculation without the extra play. The model produces a rich set of hypotheses whose expectations do not change substantially by including additional moves.

4. Again, a more realistic assumption would involve a continuum of player types. The LCT and HCT type dichotomy is a useful approach in the initial analysis. Future research extends the model by relaxing this assumption.

THE MODEL

The two-sided incomplete information game is represented in extensive form in Figures 1a and 1b. I begin by presenting a set of assumptions and then discuss how the model is used to identify the escalation strategies players adopt in equilibrium.

Prior to the game, nature assigns a cost tolerance type to each player. Recall that an LCT actor will withstand one high-escalation move to achieve a demand, whereas an HCT actor will tolerate two high-escalation moves. Thus LCT and HCT types are distinguished on the basis of their preference orderings over the outcomes. An LCT i prefers $g_i > h_i > d_i > f_i > b_i > c_i > e_i > a_i$. An HCT i prefers $g_i > c_i > e_i > h_i > a_i > f_i > b_i > c_i$.[5] Nature informs players of their type, and each player's type remains private information. Each player knows the probability that his or her opponent is an LCT or HCT type. In Figures 1a and 1b, LCT i and j types are represented by p and s, respectively. HCT i and j types are denoted $1 - p$ and $1 - s$, respectively. Because there are only two player types, $p + (1 - p) = 1.0$ and $s + (1 - s) = 1.0$. The limiting case occurs when p (s) assumes a value of 0 or 1. In either or both of these circumstances, j (i) knows the opponent's cost tolerance. For the remainder of the discussion, I assume that $0 < p$ and $s < 1$.

The players move sequentially and choose between escalating low or high. The combination of the players' three choices and the different mixes of cost tolerant actors produces 32 outcomes in the game, only 16 of which can occur given i's type. The players' payoffs for each outcome are represented generically by letters, and the payoffs are common knowledge. To clarify the discussion, consider the case in Figure 1a where an LCT i has observed two high-escalation moves. Because LCT players are willing to suffer only one high-escalation move, i escalates low on his or her final move and receives the payoff l_i. (Player i knows j is HCT after observing two high-escalation moves.) Given the same sequence of moves, an HCT i will escalate high and receive the payoff k_i.

Player i's decision to escalate low or high on his or her final move implies that i does not concede to j's demand. To put closure on the model, I assume that escalating low on the last move is equivalent to zero cost imposition (i.e.,

5. Given these preference orderings, an LCT i will have greater expected utility for the outcomes associated with escalating low on his or her final move, except when both i and j escalate low on their first move. In this case, i escalates high (j concedes) and i receives payoff g_i. The payoffs a through h and k, m, n, r, t, v, and y indicate that i is playing against an LCT or HCT j, respectively. I assume that i's expected utility for $k_i = a_i$, for $m_i = b_i$, for $n_i = c_i$, and so on. Thus i's expected utility does not change because of j's type. An LCT j prefers $f_j > h_j > d_j > g_j > b_j > e_j > c_j > a_j$. An HCT j prefers $f_j > b_j > h_j > d_j > g_j > e_j > c_j > a_j$.

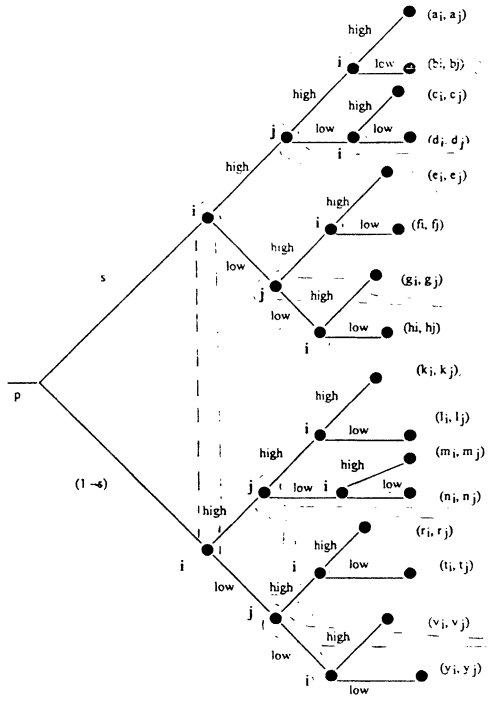


Figure 1a: Two-Sided Incomplete Information: A Lower Cost Tolerant *i* versus a Lower Cost Tolerant or Higher Cost Tolerant *j*

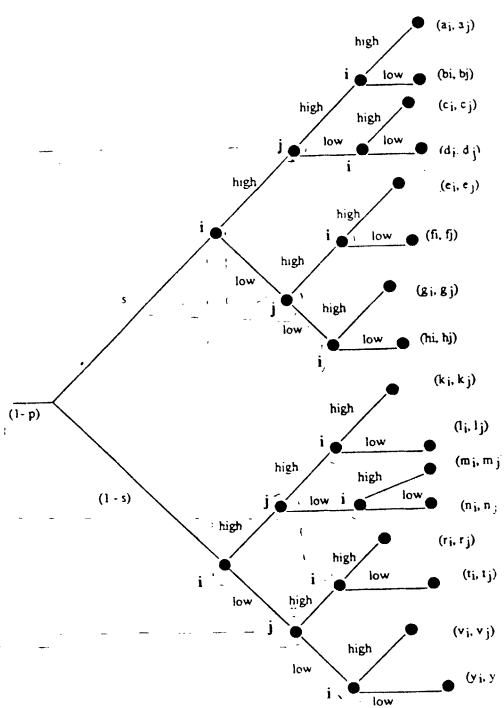


Figure 1b: Two-Sided Incomplete Information: A Higher Cost Tolerant *i* versus a Lower Cost Tolerant or Higher Cost Tolerant *j*

i concedes to j's demand). This assumption holds only for i's final move. If i escalates high on the last move, then I assume that j concedes to i's demand. Finally, if the players escalate high (or low) at each decision node, then I assume that nature ends the game by imposing a solution that lies midway between the players' demands.

Player i begins the game knowing that j is LCT with probability s and HCT with probability $1 - s$. Because neither escalation option gives i a certain payoff, i must consider a set of gambles in which one of n outcomes can occur. I assume that i's preferences among these gambles are represented by von Neumann-Morgenstern utility functions and that i chooses the level of escalation that maximizes his or her expected utility. I also assume that i is risk neutral.[6]

I argued previously that neither player type has a dominant strategy to escalate low or high. Because both j types play low or high escalation with some positive probability, i's best response to j's strategy is to play a mixed strategy. The mixed strategy tells i to play high escalation with probability x and low escalation with probability $1 - x$.

Player i uses j's payoffs to calculate the mixed strategy that makes j indifferent between his or her two strategy choices. Player i knows the preference ordering associated with each j type, but i remains uncertain of j's true type. Thus i calculates two mixed strategies: one used against LCT js (escalate high with probability x' and escalate low with probability $1 - x'$) and another mixed strategy used against HCT js (escalate high with x'' and escalate low with $1 - x''$).

However, i cannot play either mixed strategy with a 1.0 probability because j's true type remains private information. Player i randomizes between the two mixed strategies using the probabilities that j is an LCT or HCT type. This overall mixed strategy tells i to escalate high with probability x and escalate low with probability $1 - x$. This strategy maximizes i's expected utility by making both an LCT and an HCT j indifferent between escalating low or high.

The strategies that players adopt in equilibrium are identified using the perfect Bayesian solution concept. This equilibrium is a strategy combination and a set of beliefs that satisfies two conditions. First, at each decision node in the game, the strategies for the remainder of the game are Nash given the beliefs and strategies of the other players. Second, the players' beliefs at each information set are rational given the evidence appearing that far in the game (Rasmusen 1989, 110). Player i's equilibrium strategy specifies that i play a mixed strategy that makes both j types indifferent between escalating low or

6. The assumptions made about player i also pertain to player j.

high. An LCT i determines his or her mixed strategy according to the following:

$$ps*[(x)*(b) + (1-x)*(f)] + p(1-s)*[(x)*(l) + (1-x)*(y)] =$$
$$ps*[(x)*(e) + (1-x)*(g)] + p(1-s)*[(x)*(n) + (1-x)*(t)].$$

Isolating x, the probability that an LCT i escalates high, produces equation 1.1:[7]

$$x = \frac{p[(s*(g-t-f+y) + (t-y)]}{p*[(s*(b-f-l+y-e+g+n-t) + (l-y-n+t)]}. \tag{1.1}$$

In equation 1.1, the dependent variable, x, is the probability that an LCT i escalates high. The independent variables are located on the right-hand side of the equation. The variables reflect j's payoffs for different outcomes in the game and both players' prior estimates of the opponent's type. By manipulating the values of the independent variables, equation 1.1 produces a plethora of hypotheses identifying different conditions under which an LCT i escalates low or high. Additional hypotheses can also be generated using an equation reflecting the equilibrium mixed strategy for an HCT i. A sample of the hypotheses derived from equation 1.1 are listed here. Several of the hypotheses appear intuitive; however, these interrelationships become rather complex once they are operationalized.

Hypothesis 1: An LCT i is more likely to escalate high when s, the probability that j is LCT, decreases.

Hypothesis 2: An LCT i is more likely to escalate high as the difference between the value of winning and losing to j increases.

Hypothesis 3: An LCT i is more likely to escalate high as the expected utility of escalating low for either LCT or HCT j actors decreases.

The first hypothesis is tested empirically and then is discussed shortly. First, I expose the logic behind the second hypothesis and then elaborate on j's role in the game. The second hypothesis is derived by increasing j's payoffs for either b_j, d_j, or f_j while holding the remaining variables constant in equation 1.1. The same hypothesis can be obtained by decreasing j's payoffs for c_j, e_j, or g_j while holding all other variables constant. The second hypothesis identifies the conditions under which an LCT i perceives j to be HCT. Player i's perception may be a function of the issue at stake such as j's effort to build a state (e.g., Serbia). Because j will tolerate a great deal of cost before

7. See Appendix A for the mathematical derivation of this equation.

conceding to i's demand, i can maximize his or her expected utility by concealing his or her LCT type and escalating high at the onset of the dispute.

Player j, who moves second in the game, has the opportunity to observe i's escalation and to revise his or her prior beliefs concerning i's type. Player j has two pieces of information. First, j knows the prior probability that i is an LCT or HCT type. Second, j is aware that neither i type has a dominant strategy to escalate low or high. Assume that j observes high escalation. Player j uses Bayes's rule to calculate the posterior probability that the player escalating high is also LCT:

$$\frac{P(ple)*(p)}{P(ple)*(p) + P[(1-p)le]*(1-p)}$$

where p = the prior probability that i is LCT, $(1-p)$ = the prior probability that i is HCT, $P(ple)$ = the probability that an LCT i escalates high, and $P[(1-p)le]$ = the probability that an HCT i escalates high. Player j uses his or her revised estimates of i's type to assess whether the outcomes associated with low or high escalation provide j with greater expected utility.

$$\left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(bj) + 1 - \left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(kj) =$$

$$\left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(dj) + 1 - \left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(mj)$$

where p = prior probability that i is LCT, $(1-p)$ = prior probability that i is HCT, $P(plx)$ = probability that an LCT i escalates high (call this w), $P[(1-p)ly]$ = probability that an HCT escalates high (call this z), b_j = the value to j of i's concession minus two high-escalation costs (i is LCT), k_j = the value to j of the nature's imposed compromise minus three high-escalation costs (i is HCT), d_j = the value to j of conceding to i minus one high-escalation cost, and m_j = the value to j of conceding to i minus two high-escalation costs.

Isolating w,[8]

$$w = \frac{(m-k)}{p*(b-k-d+m)}. \tag{1.2}$$

The term w reflects j's updated probability that the player escalating high is LCT; $1-w$ is the updated probability that the player escalating high is HCT. Equation 1.2 also produces a rich set of hypotheses that identify the

8. See Appendix B for the mathematical derivation of this equation.

conditions under which *j* types escalate low or high. One of the hypotheses states that as the value of *w* increases (*i* is more likely to be LCT), an HCT *j* is more likely to escalate low. The theory states that HCT *j*s are in a stronger position to secure the opponent's concession more cheaply than are LCT actors. If low escalation fails to convince *i* to concede, then HCT *j*s are willing to escalate to higher levels.

This completes a description of the game. The first hypothesis derived from equation 1.1 is discussed and then is tested empirically in the subsequent section. This hypothesis states that an LCT *i* is more likely to escalate high when *s*, the probability that *j* is LCT, decreases.

The hypothesis leads to the expectation that LCT types are more likely to escalate high as the disparity between the players' cost tolerances increases. The intuition behind this hypothesis is illustrated by the Japanese attack on Pearl Harbor in 1941. The Japanese high command calculated that their military had, at best, 4 to 5 months to solidify gains in South Asia and the Pacific before resources would be exhausted. To accomplish this goal, the U.S. Navy had to be rendered incapable of interfering with the Japanese mission. The one thing of which the Japanese were fairly certain was that America's greater resources would prevent the Japanese from prevailing in a war of attrition. The Japanese were willing to gamble that as a result of their escalating high from the outset, the United States would come to terms rather than continue the escalation game.

Stated more generally, LCT actors are less likely to obtain an opponent's concession by playing a game of low escalation. Thus LCT actors must behave as if they were HCT. One way to accomplish this is by demonstrating a willingness to suffer higher escalation costs early on in the game. If high escalation fails to convince *j* to concede, then LCT actors are likely to give in.

OPERATIONAL HYPOTHESIS

The dependent variable of the hypothesis is the probability that the first escalation action taken by the LCT actor will also be the highest level of escalation that actor will achieve in the dispute. Operationalizing the dependent variable requires comparing the level of escalation associated with two actions. A criterion is established to determine whether a subsequent action was taken at the same, a lower, or a higher level of escalation relative to the first action in the dispute. There have been a number of attempts to categorize different escalation actions (Blechman and Kaplan 1978; Gochman and Maoz 1984). These efforts guide the discussion here.

I assume that different escalation actions can be distinguished from one

another in terms of the costs borne by an actor for carrying out an action. Certain groups of escalation tactics are considered more or less costly than others based on the current costs that a player must pay to execute the action. Clearly, the lowest point on the escalation continuum is represented by no action or zero cost imposition. An actor who fails to escalate is effectively choosing to withdraw from the escalation game. The second least costly option is a threat to use force. An actor experiences relatively few immediate costs in proclaiming that he or she will engage in some future coercive behavior. Many of the costs suffered by the threatening actor will be paid at a later date either because the threat was actually carried out or because the actor incurs reputation costs for failing to follow through on a previous threat of escalation. A show of force is an implicit threat to use force. Yet a show or display of force is considered a higher escalation level than is a threat given that there are immediate costs associated with mobilizing troops and the like. A use of force is considered the next highest level of escalation. These escalation tactics subsume the acts of demonstrating force (and their attendant costs) and generate additional costs by directly engaging the opponent's resources in hostile actions. Finally, the maximum level of escalation is war.

The independent variable of the hypothesis is the relative value of the players' cost tolerances. Recall that cost tolerance is defined as the maximum costs an actor is willing to suffer to secure some value in conflict. Achieving a perfect correspondence between the conceptual and operational meanings of cost tolerance would require determining how an actor evaluated an issue in a dispute for all disputes included in the analysis. The limitations associated with this exercise are rather serious; however, by stressing a state's ability to impose escalation costs, it is possible to achieve a fairly reasonable surrogate for cost tolerance.

Military capabilities are among the most important tools states have to impose costs on their adversary (Organski and Kugler 1980; Small and Singer 1982). All other things being equal, an increase in a player's relative military capabilities should also be associated with an increase in that player's cost tolerance. As Rosen (1972) notes, the per-unit cost of using force decreases as a player's military capabilities increase. In addition, the ability of a relatively stronger actor to replenish expended resources more efficiently also suggests that cost per unit of force is declining due to economies of scale (Cannizzo 1980). It is also reasonable to suspect that states that are willing to consume the costs associated with developing stronger military capabilities are likely to be heavily involved in the issues of the international system and may feel that they have a stake in preserving the status quo (Organski 1968).

The argument suggests that there is an association between cost tolerance

and military capabilities. Clearly, these variables do not overlap completely; however, military capabilities serve as a reasonable approximation of cost tolerance. The operational hypothesis leads to the expectation that as the disparity in the actors' military capabilities increases, the probability that the weaker actor escalates to a higher level subsequent to the first escalation action decreases. In the following section, the research design is presented, followed by an analysis of the results of the empirical test.

RESEARCH DESIGN

The hypothesis is tested in the context of militarized interstate disputes (MIDs) (Gochman and Maoz 1984). MIDs are defined as a

> set of interactions between or among states involving threats to use military force, displays of military force or actual uses of military force. To be included, these acts must be explicit, overt, non-accidental and government sanctioned. (p. 587)

The nature of MID crises means that there is an increase in the probability that one or both of the players will resort to the use of force to resolve the conflict. Consequently, MIDs provide a rich context for analyzing the theoretical expectations of the model.

An appropriate test of the theory requires that the dependent variable capture different levels of escalation. The militarized interstate incident dispute data set (Gochman and Maoz 1984) identifies the kinds of actions and the number of incidents (actions) taken by each actor in a dispute. State actions are represented in the incident data set by a variable called a military confrontation action (MCA). There are 14 different MCAs that are grouped into four escalation categories: threats of force, displays of force, uses of force, and war. According to the argument presented earlier, the highest level of escalation is war, followed by a use of force, a display of force, and a threat to use force.[9]

The dependent variable is the escalation behavior of the LCT actor. The expectation is that the LCT actor will achieve his or her highest escalation level of the entire dispute on his or her first move. Thus the LCT actor's first action (and its appropriate escalation level) must be compared to the escalation level of any subsequent action in the dispute. A dichotomous dependent variable was created to identify cases in which an LCT actor escalated at the same or to a higher level. The dependent variable was scored 1 if the first and

9. The "no action or no escalation" category is not represented in the data set. Therefore, the lowest level of escalation action is a threat to use force.

highest actions in the dispute represented the same level of escalation or 0 if the LCT actor escalated to a higher level.

The independent variable, cost tolerance, was operationalized in terms of military capabilities. The capabilities data were obtained from the Correlates of War (COW) data set (Small and Singer 1982). Military capabilities are measured using six different indicators: military expenditures, military personnel, iron/steel production, energy consumption, total population, and urban population. These indicators are combined in the COW data set to create an index reflecting a state's percentage of the total capabilities in the world for each year. From this index, I created a variable reflecting the ratio of military capabilities of the actors per dispute. The stronger state is represented in the numerator and the weaker state in the denominator. The ratio can vary from 1 (the actors' capabilities are equal) to any positive number.

Because the hypothesis refers to dyadic disputes, any cases in which nations joined an ongoing dyadic dispute were excluded from the analysis. In addition, all minor-minor disputes had to be excluded for lack of data. With these exclusions, the test was performed on 119 cases. The cases included the years from 1816 to 1976. Before proceeding with the data analysis, it is important to stress that the hypothesis is tested at the monadic level of analysis. The hypothesis refers exclusively to the escalation behavior of the LCT (militarily weaker) actor. The results of the empirical test are presented in the following section.

DATA ANALYSIS

The hypothesis was tested using logit. There are 39 cases in which the weaker side escalated to a higher level and 80 cases in which the first and highest levels of escalation were the same. The estimated equation includes a control variable that identifies the first level of escalation taken by the weaker side in the dispute. The control variable is not an explanatory variable because the theory does not predict the initial level of escalation. The theory does state that, given the first action (whatever that action might be), the weaker nation is less likely to escalate to higher levels in the dispute. Excluding this variable could potentially bias the findings in favor of rejecting the null hypothesis. The reason is that states are constrained from escalating to higher levels depending on the initial level of escalation. For instance, a state that uses force from the outset can achieve a higher level of escalation only by going to war, whereas a state that starts the dispute with a

threat to use force can escalate to three higher levels. The test results are presented in Table 1.

The test results are consistent with the expectation of an inverse relationship between cost tolerance and LCT actors' escalation behavior. This finding is particularly encouraging in lieu of the fact that there is a likely selection bias in the test cases. The selection bias is a result of the imperfect correspondence between cost tolerance and its surrogate measure, military capabilities. The bias is against those cases in which less capable actors were excluded from the data set because they did not place sufficient value on their demand to reach the crisis stage. Therefore, the less capable actors that are included in the analysis are likely to have higher cost tolerances relative to their excluded, less capable counterparts. This selection bias provides for a tougher test of the theory by making it more difficult to reject the null hypothesis. With this in mind, the test results indicate clearly that an increase in the disparity between the players' cost tolerances is associated with a decrease in the probability that LCT actors escalate to higher levels. I also expected that the relationship between the variables would be modest given the association between military capabilities and cost tolerance. This relationship is also confirmed by the test results.

The naive model, which simply predicts the outcome based on the modal category, predicts 67% of the cases correctly. By comparison, the logit model reduces the frequency of incorrect predictions by 12%. The reduction in error may appear modest, but note that the distribution of cases on the dependent variable allows the naive model to perform quite well in the absence of any information on the actors' relative cost tolerances. The explanatory variable must be able to discriminate strongly between the cases to generate any substantial improvement in the frequency of accurate predictions. The military capabilities variable is distinguishing between the cases; however, for the reasons already mentioned, the effect of this variable is not particularly strong. This conclusion is corroborated noting how different values for the relative capabilities variable affect the probability of observing weaker actors escalating to higher levels.

Relative to conditions of rough military equality, actors who are 100 times weaker than the opponent are 13% less likely to escalate to higher levels. The large change in moving from approximate parity to gross power differentials produces a fairly small change in the probability that weaker actors fail to achieve higher escalation levels. The interpretation of the logit coefficients confirms the expectation that the capabilities variable has difficulty in discriminating strongly between the cases.

Contingency tables are useful diagnostic tools for identifying the percent-

TABLE 1
Logit Estimates for Weaker State's Escalation Behavior ($N = 119$)

Variable	*Coefficient*	*ΔProbability*
Intercept	4.5139	—
	(1.03)	
Ratio	–.006	.01
	(.004)	
First escalating action	–1.63*	—
	(.333)	

NOTE: Numbers in parentheses are standard errors of the estimates. Percentages predicted correctly: null = 67%, logit = 78%.
*$p < .0001$.

age of cases that fall within the predicted categories. With this analysis, different levels of cost tolerance are compared to the escalation behavior of weaker actors within each category. The contingency table is presented in Table 2.

The columns represent four different levels of relative military capabilities. The values in each column were obtained using the interquartile range. The rows reflect the escalation behavior of LCT actors. No escalation represents cases in which LCT actors did not escalate to higher levels. The second and third rows are cases in which LCT actors escalated one or two levels higher, respectively. The expectation is that the percentage of cases falling within the no-escalation row will decrease moving left to right across the capabilities columns. Similarly, I expect that there will be an increase in the number of cases in the second and third rows, moving left to right.

In the no-escalation row, weaker actors, who were roughly equal to twice as weak as their opponent, failed to escalate to higher levels 72% of the time. The first increase in the number of no-escalation cases above 72% occurs when LCT actors are roughly 60 times weaker. I expected that the number of cases would increase consistently across all of the power categories. This expectation holds for slight and huge inequities in military capabilities. An examination of cases in which weaker actors escalated one or two levels higher reveals a similar pattern. The percentage of cases in the fourth column decreases relative to that of the first column for both escalation levels. This is also true for cases in which weaker actors escalate two levels higher and were 2 to 12 times weaker than the opponent.

As mentioned previously, I assume that there is an association, although

TABLE 2
Level of Escalation Achieved by
the Weaker State in a Militarized Interstate Dispute

	Power Ratio[a]			
	Ratio < 2.44 (N = 29)	*Ratio ≥ 2.44 and < 12.03* (N = 30)	*Ratio ≥ 12.03 and < 59.31* (N = 30)	*Ratio ≥ 59.31* (N = 30)
Escalation level				
No escalation	21	19	15	25
	(72)	(63)	(50)	(83)
One level higher	4	8	10	3
	(14)	(27)	(33)	(10)
Two levels higher	4	3	5	2
	(14)	(10)	(17)	(7)

NOTE: Percentages are in parentheses.
a. Power ratio = disparity in military capabilities of stronger state over weaker state (e.g., a power ratio of 2 means that stronger state is twice as strong as the opponent).

not a strong relationship, between military capabilities and cost tolerance. The contingency table analysis reveals that the association between the variables may not be consistent across all levels of power disparities.[10] The theory does not identify the factors that may be contributing to the uneven results across the set of dispute cases. Further investigation is required to determine the nature of this relationship.

Despite this empirical curiosity, the test results confirm the theory's expectations. It is clear that as the disparity in the players' cost tolerances increases, LCT actors are less likely to escalate to higher levels relative to their first actions in the dispute. An LCT actor is aware that his or her cost tolerance will be surpassed before that of an HCT adversary by playing a game of low escalation. To have any chance of winning the opponent's concession, LCT actors must behave as if they are willing to bear greater escalation costs. One way for LCT actors to accomplish this is to escalate to the maximum of their ability at the onset of the dispute.

10. The percentages of cases falling within all cells in the third column (a 12:1 to 60:1 power ratio) are all inverse of the model's theoretical expectations. This suggests that there may be a systematic pattern to the cases in this category. An analysis of the cases in this column revealed that 25% of the cases were drawn from World War II. This raises the possibility that the cases are not independent due to a complex set of alliance patterns. A logit analysis was performed excluding the World War II cases. The results were consistent with the first empirical test.

CONCLUSION

Escalation occupies a central place in many studies of conflict resolution processes. To date, however, there have been few attempts to develop a general theory of escalation aimed at deriving a set of testable hypotheses that hold across different contexts. One way of improving our understanding of conflict processes and outcomes is to focus our attention on escalation processes in general and to subject hypotheses to empirical tests across different contexts. One of the major concerns dominating the crisis behavior research has been to identify the conditions under which states escalate. Another important question, and one that has received less systematic theoretical or empirical treatment, is the level of escalation that states are likely to achieve once the decision to escalate has been made. This article addresses that gap by presenting a general theory of escalation that produces a rich set of hypotheses identifying the levels of escalation states will achieve in conflict.

One hypothesis leads to the expectation that as the disparity in the players' cost tolerances increases, the probability that LCT actors escalate to higher levels relative to the first dispute actions decreases. An LCT actor is less likely to secure an HCT player's concession using a series of low-escalation actions. By escalating to the maximum from the onset of the conflict, an LCT actor can demonstrate a willingness to suffer the costs of escalation. Thus an LCT actor is motivated to behave as if he or she has higher cost tolerance than is actually the case. The test results confirm the theory's expectations, and this is especially encouraging given the imperfect correspondence between military capabilities and cost tolerance.

This finding is consistent with the conclusions of many formal models seeking to explain why weaker actors do not capitulate immediately to their stronger opponent (Morrow 1989; Lalman 1990; Zagare 1992). The theory explains why the level of escalation observed depends on the players' beliefs about their environment and their opponent's willingness to suffer escalation costs. Frequently, discussions pertaining to states' escalation strategies assume that some optimal condition or level of escalation exists independent of the strategic context (Ikle 1971; Lockhart 1979; Haig 1984).

Moreover, the findings indicate clearly that the theory merits further investigation. One of the model's hypotheses was tested in this article, but many other hypotheses could be tested to provide a more comprehensive picture of escalation processes. However, a more complete understanding of escalation processes can be achieved only when our thinking is guided by a systematic, rigorous theory. Escalation can be too dangerous and costly for society when guided soley by policymakers' intuition.

APPENDIX A
A Lower Cost Tolerant *i*'s Mixed Strategy

1. $ps[(x)^*(b) + (1-x)^*(f)] + p(1-s)[(x)^*(l) + (1-x)^*(y)] =$
   $ps[(x)^*(e) + (1-x)^*(g)] + p(1-s)[(x)^*(n) + (1-x)^*(t)]$
2. $ps(bx + f + fx) + (p - ps)(lx + y - yx) = ps(ex + g - gx) + (p - ps)(nx + t - tx)$
3. $psbx + psf - psfx + plx + py - pyx - pslx - psy + psyx =$
   $psex + psg - psgx + pnx + pt - ptx - psnx - pst + pstx$
4. $(psbx - psfx - pslx + psyx) + (plx - pyx) + (psf - psy) + py =$
   $(psex - psgx - psnx + pstx) + (pnx - ptx) + (psg - pst) + pt$
5. $(psbx - psfx - pslx + psyx - psex + psgx + psnx - pstx) + (plx - pyx - pnx + ptx) =$
   $(psg - pst - psf + psy) + (pt - py)$
6. $psx\,(b - f - l + y - e + g + n - t) + px(l - y - n + t) = ps(g - t - f + y) + p(t - y)$
7. $px[(s^*(b - f - l + y - e + g + n - t) + (l - y - n + t)] = p[(s^*(g - t - f + y) + (t - y)]$
8. $$x = \frac{p[(s^*(g - t - f + y) + (t - y)]}{p[(s^*(b - f - l + y - e + g + n - t) + (l - y - n + t)]}$$

NOTE: Subscripts on player *j*'s payoffs—b, f, l, y, e, g, n, and t—are dropped.

APPENDIX B
Player *j*'s Revised Estimates of *i*'s Type

$$\left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(bj) + 1 - \left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(kj) =$$

$$\left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(dj) + 1 - \left[\frac{P(ple)*(p)}{P(ple)*p + P[(1-p)le]*(1-p)}\right]*(mj)$$

$$\left[\frac{wp}{wp + z(1-p)}\right]*(\mathrm{b}) + 1 - \left[\frac{wp}{wp + z(1-p)}\right]*(\mathrm{k}) =$$

$$\left[\frac{wp}{wp + z(1-p)}\right]*(\mathrm{d}) + 1 - \left[\frac{wp}{wp + z(1-p)}\right]*(\mathrm{m}) =$$

$$\left[\frac{wpb + k - wpk}{wp + z(1-p)}\right] = \left[\frac{wpb + m - wpm}{wp + z(1-p)}\right]$$

$(wpb + k - wpk)*[(wp + z(1-p)] = (wpd + m - wpm)*[(wp + z(1-p)]$

1. dividing both sides by $wp + z(1-p)$,
   $wpb + k - wpk = wpd + m - wpm$
2. collecting terms and subtracting,
   $(wpb - wpk - wpd + wpm) = (m - k)$
3. taking out common terms,
   $wp\,(b - k - d + m) = (m - k)$
4. dividing by $p*(b - k - d + m)$,

$$w = \frac{(m-k)}{p*(b-k-d+m)}$$

NOTE: $P(ple)$ = probability that high-escalating player is low cost tolerant (call this w). $P[(1-p)le]$ = probability that high-escalating player is high cost tolerant (call this z). Subscripts are dropped after equation.

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